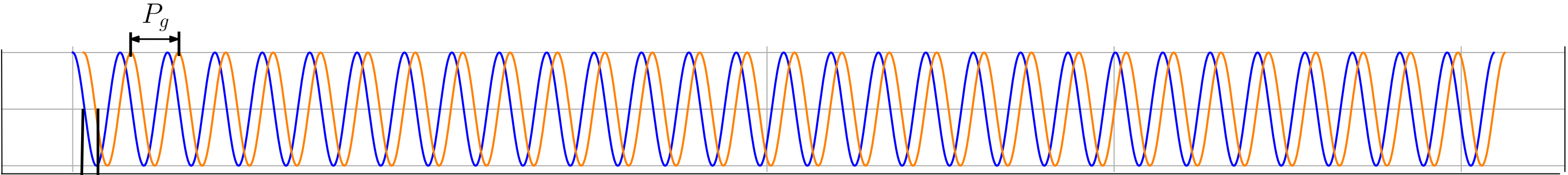


Sampling Moiré Magnification of Resolution

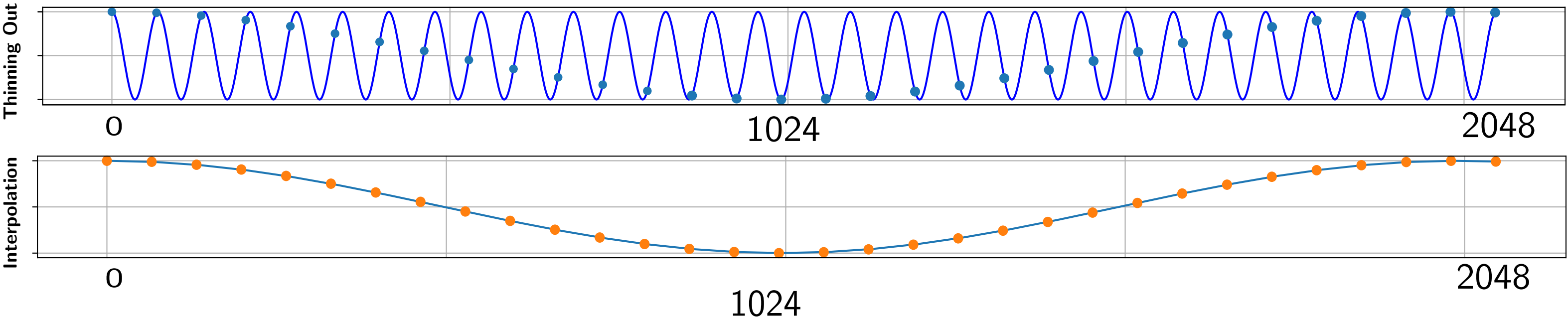


Phase Resolution, $\delta\phi = \frac{2\pi}{T}$

Displacement Resolution $\delta d = \frac{P_g}{T}$

Displacement $d = \frac{P_g \cdot \phi}{2\pi}$

T - Number of Pixels in One Cycle
N - Thinning Out Factor
 P_g - Grating Pitch
 d - Displacement Measured

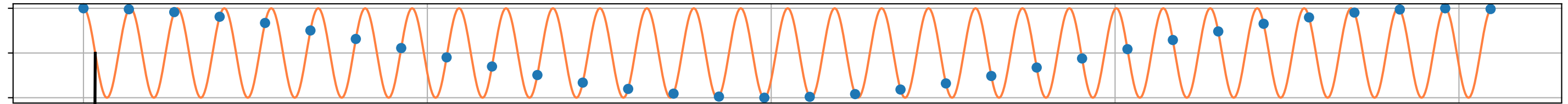


Pitch, of Generated Pattern, $M = \frac{T \cdot N}{|T - N|}$

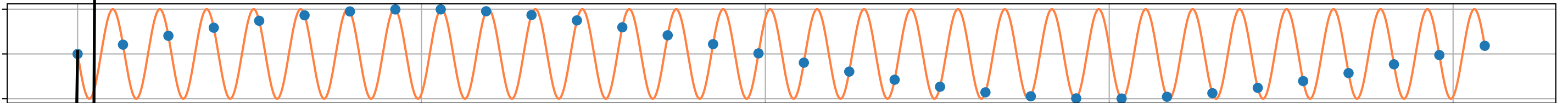
Displacement Resolution $= \frac{P_g}{T \cdot M}$

Phase Difference in Grating = Phase Difference in Generated Moiré Fringe

Reference Signal

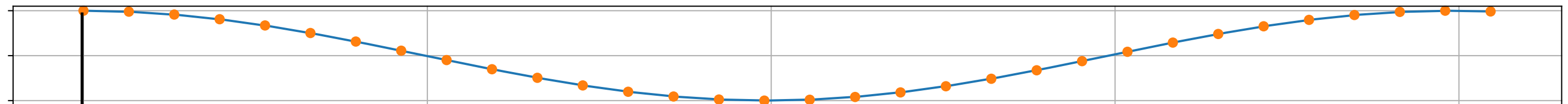


Displaced Signal with $\frac{P}{4}$ displacement

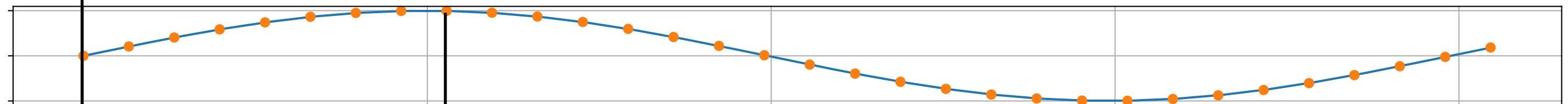


$$\Delta\phi = \frac{\pi}{2}$$

Moiré' Pattern from Reference Signal



Moiré' Pattern from Displaced Signal with $\frac{P}{4}$ displacement

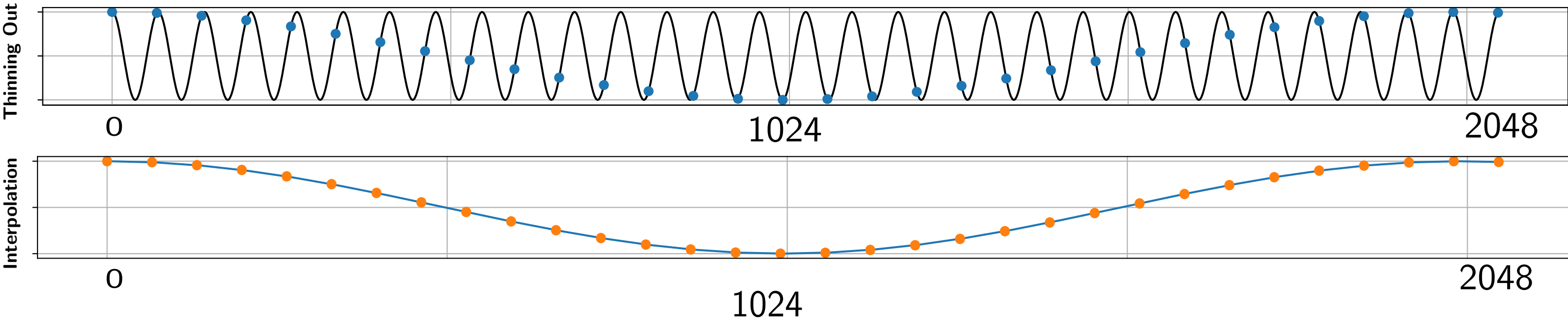
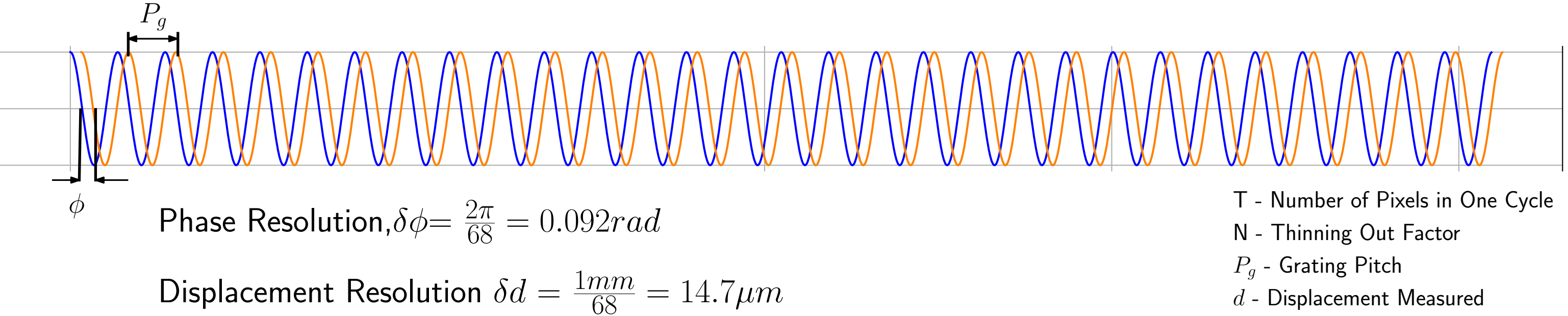


$$\Delta\phi = \frac{\pi}{2}$$

Image of Grating

Generated Moiré Fringes

Resolution Estimated



$N = 66$

Pitch, of Generated Pattern, $M = 29.1mm$

Phase Resolution = $\frac{0.092rad}{29.1} = 0.00316rad$

Displacement Resolution = $0.5\mu m$

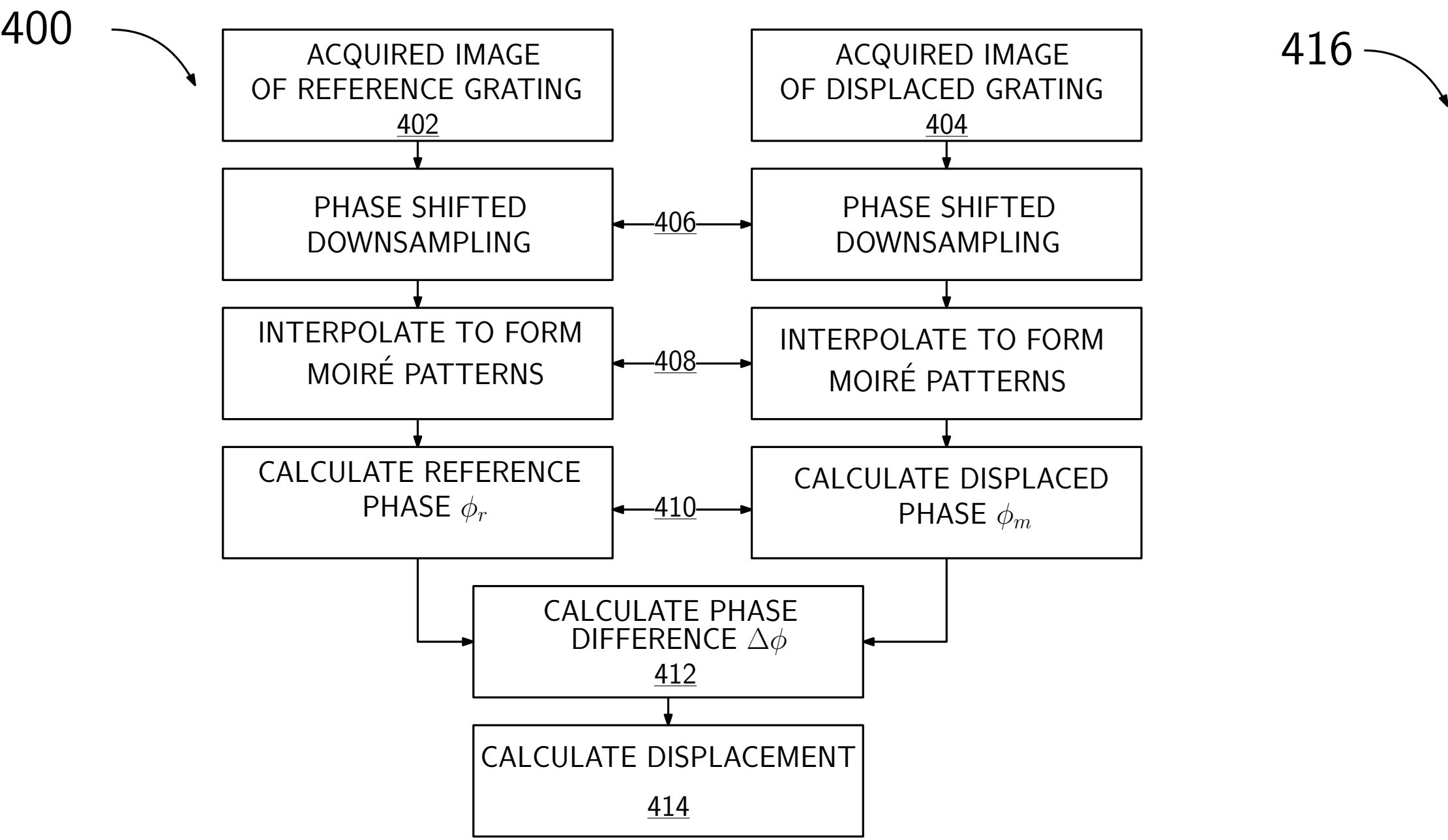


FIG. 4A

FIG. 4B

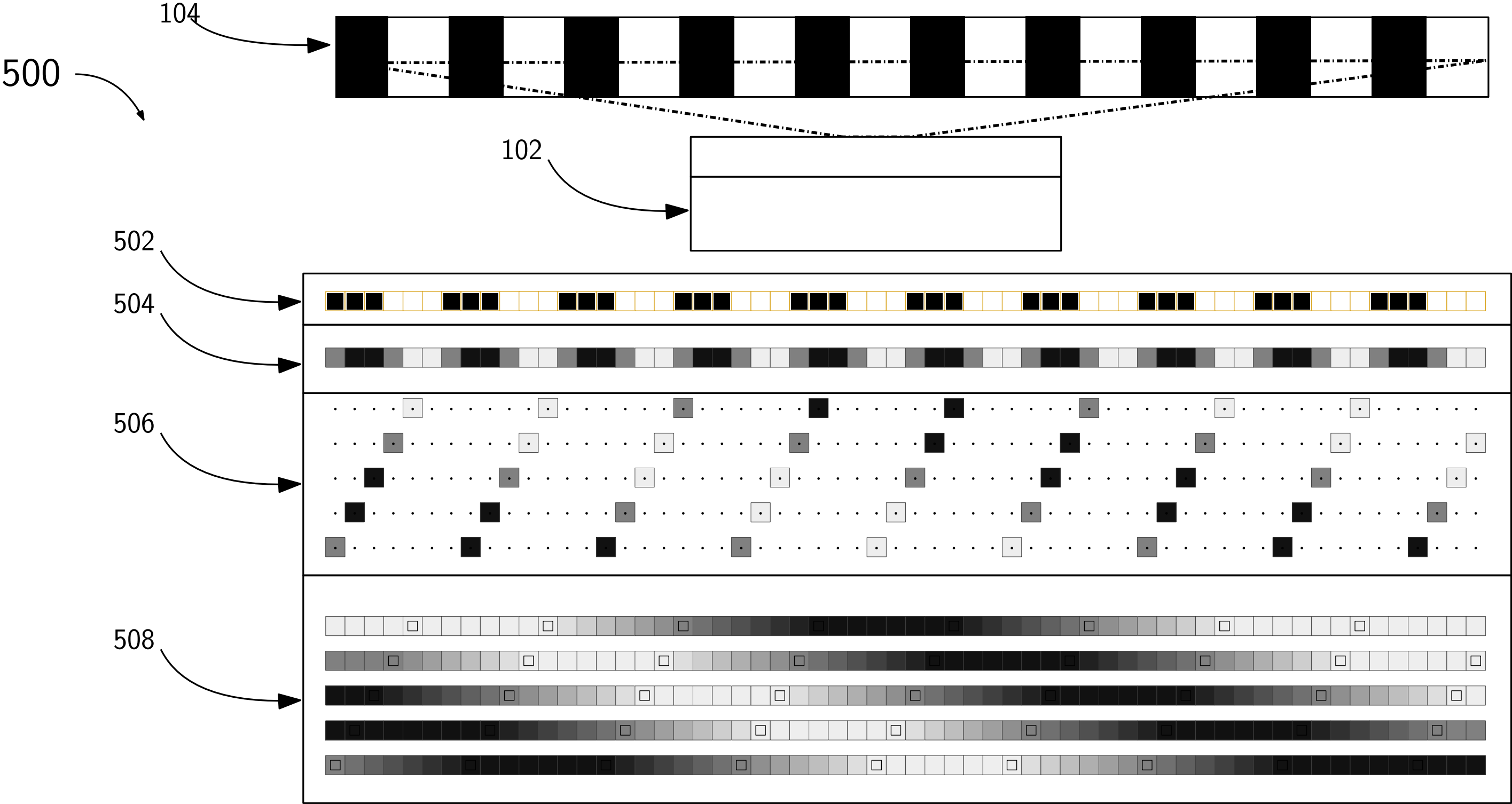
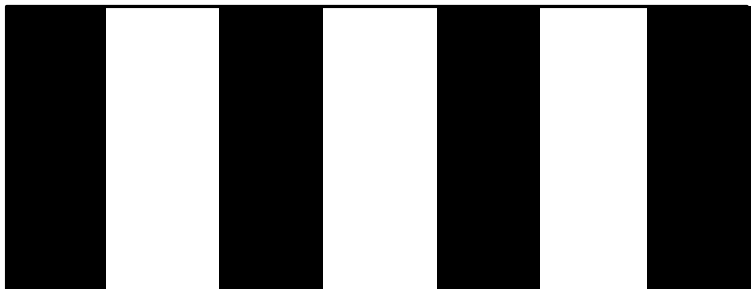


FIG. 5

700A



702

704

706

708

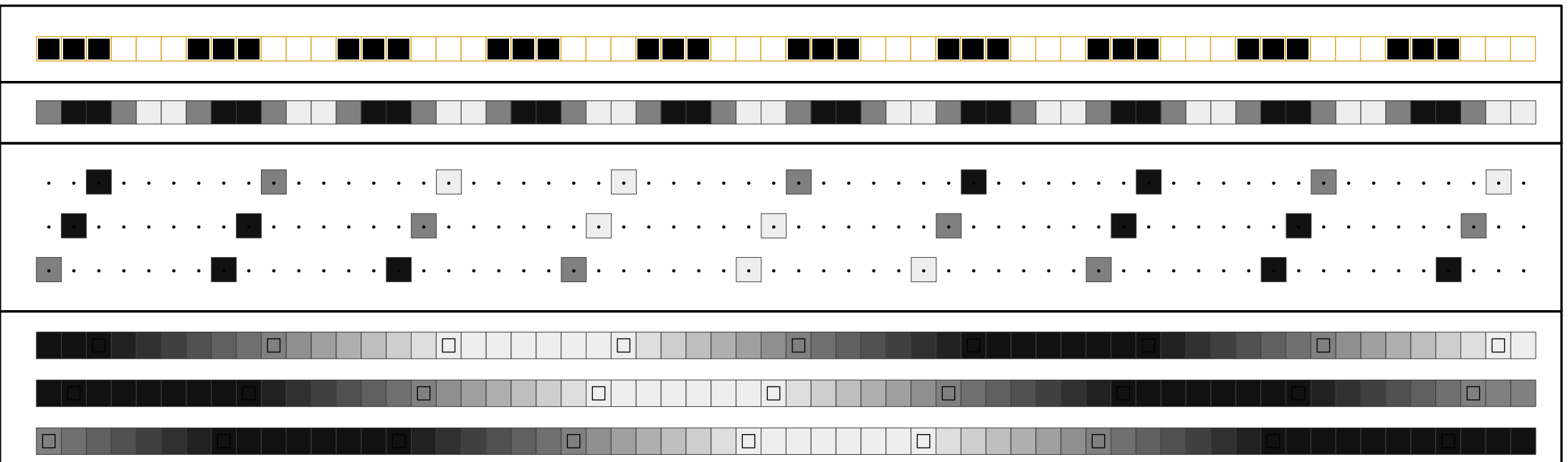
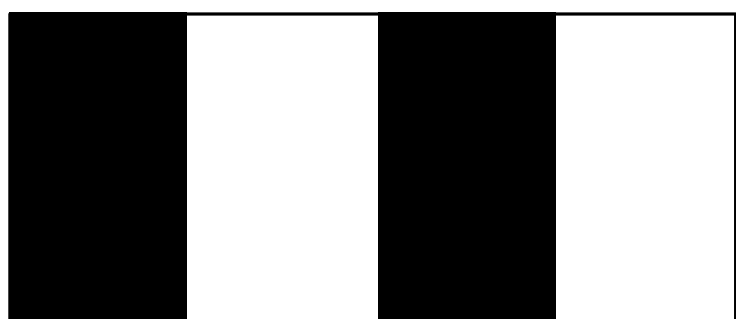


FIG. 7A

700B



710

712

714

716

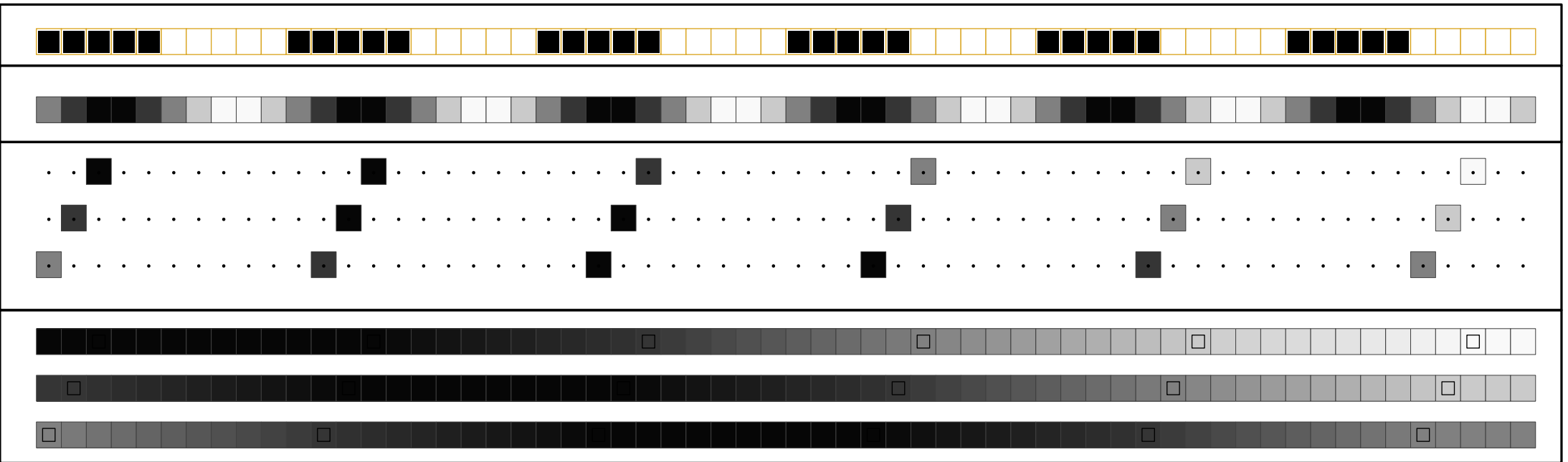


FIG. 7B